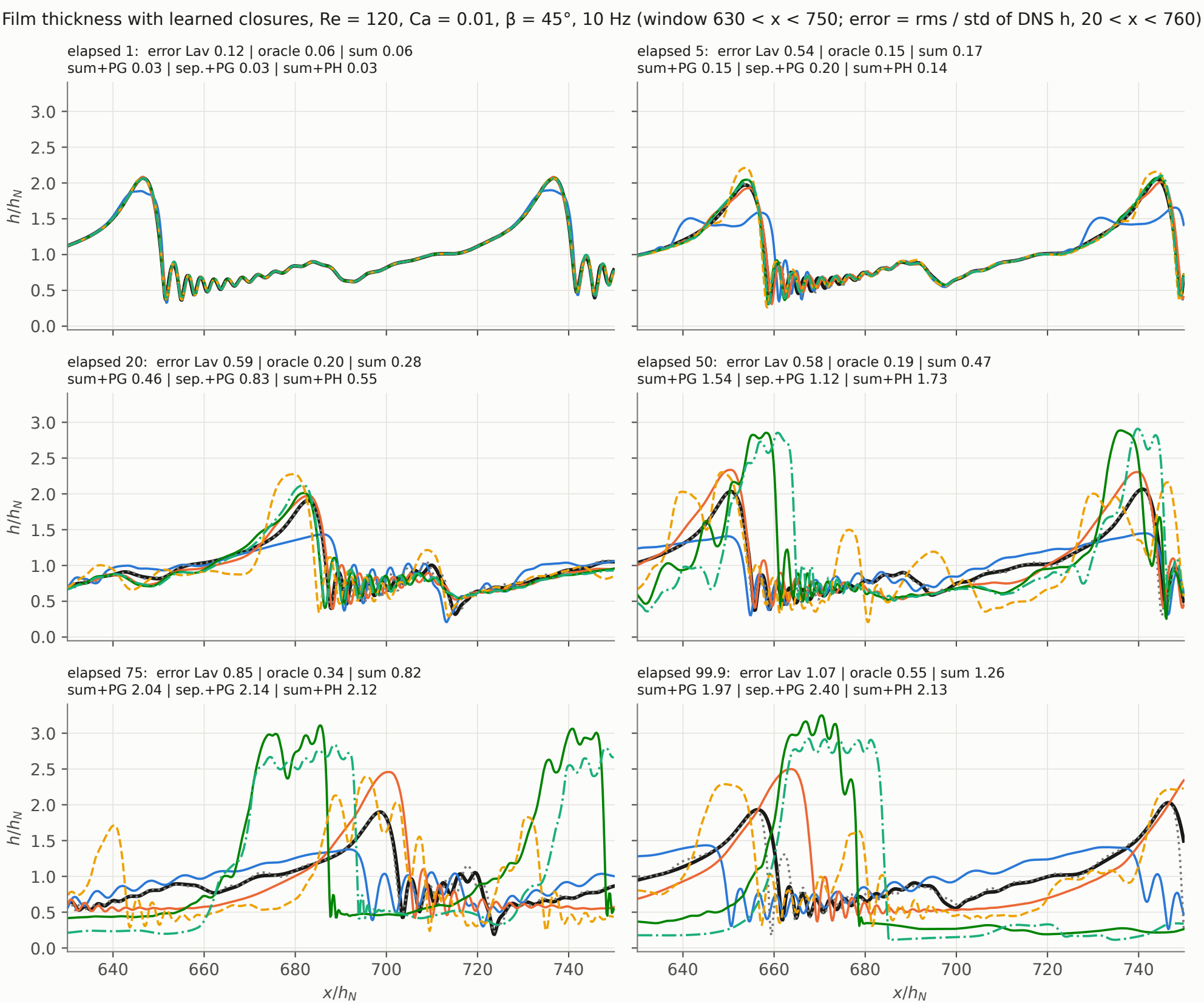

Film thickness with learned closures, Re = 120, Ca = 0.01, β = 45°, 10 Hz (window 630 < x < 750; error = rms / std of DNS h, 20 < x < 760)
elapsed 1: error Lav 0.12 | oracle 0.06 | sum 0.06
sum+PG 0.03 | sep.+PG 0.03 | sum+PH 0.03
elapsed 5: error Lav 0.54 | oracle 0.15 | sum 0.17
sum+PG 0.15 | sep.+PG 0.20 | sum+PH 0.14
elapsed 20: error Lav 0.59 | oracle 0.20 | sum 0.28
sum+PG 0.46 | sep.+PG 0.83 | sum+PH 0.55
elapsed 50: error Lav 0.58 | oracle 0.19 | sum 0.47
sum+PG 1.54 | sep.+PG 1.12 | sum+PH 1.73
elapsed 75: error Lav 0.85 | oracle 0.34 | sum 0.82
sum+PG 2.04 | sep.+PG 2.14 | sum+PH 2.12
elapsed 99.9: error Lav 1.07 | oracle 0.55 | sum 1.26
sum+PG 1.97 | sep.+PG 2.40 | sum+PH 2.13
h/h_N
x/h_N
0.0
0.5
1.0
1.5
2.0
2.5
3.0
640
660
680
700
720
740
DNS
Lavalle only
Oracle ΔT_w + ΔM_x
ML sum (h, h_x)
ML sum (h, h_x) + ML ΔP_x (G)
ML ΔM_x + ML ΔT_w + ML ΔP_x (G)
ML sum (h, h_x) + ML ΔP_x (H)